Solving Quadratics using a Calculator

Exam code: 0607

Download PDF

Written by: Jamie Wood
Reviewed by: Dan Finlay
Updated on 11 June 2024

Solving Quadratics using a Calculator

How do I solve quadratic equations using my calculator?

- Your **graphic display calculator** is able to solve quadratic equations
- Within the menu you will find an option for **solving a polynomial**
  - This may be within the **equation solver** section on some models
- Select the **degree** or **order** of polynomial you are solving
  - When solving a **quadratic**, the **degree** or **order** is **2**
- Type in the **coefficients** of the quadratic you are solving
  - This is usually in the format
    - $a_2x^2 + a_1x + a_0 = 0$ or
    - $ax^2 + bx + c = 0$
  - E.g. To solve $x^2 + 6x - 4 = 0$
    - $a_2 = 1$, $a_1 = 6$, $a_3 = -4$ or
    - $a = 1$, $b = 6$, $c = -4$
  - You must therefore **rearrange** the equation you are solving to this format (... = 0)
- Then press **enter** or **solve** to view the **solutions**

Should I round my answers?

- Your graphic display calculator will present answers in **exact form** where possible
  - E.g. $-3 + \sqrt{13}$ rather than 0.6055512755...
    - Your calculator will be able to switch between the exact form and a decimal
  - If there is **further working** to be done using this answer
    - it is best to keep it in an **exact** form
  - If the solutions to the quadratic are the **final answers** for the question

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

significant figures)

Should I show my working when using a calculator to solve a quadratic?

- In a **calculator paper** it is expected that you use your calculator to solve quadratic equations
- Therefore you do **not** need to show working (such as using the quadratic formula)
- However, you **should** write down:
  - The **equation** you are entering into your calculator (and any prior **rearranging**)
  - **Both solutions** found by your calculator
    - In exact form where possible
- This will ensure you receive as many method marks as possible, in case of an error

Examiner Tips and Tricks

Solving a quadratic on your calculator can also help factorise a quadratic:

- If the solutions to a quadratic equation are $x = -3$ and $x = 2$
- Then the equation can be written as $(x+3)(x-2) = 0$

Worked Example

Use your calculator to solve the equation $10x = 19 - 6x^2$ giving your solutions to four significant figures.

Rearrange the equation to the form $ax^2 + bx + c = 0$

$$10x = 19 - 6x^2$$
$$6x^2 + 10x = 19$$
$$6x^2 + 10x - 19 = 0$$

On your calculator, enter the menu and select the polynomial solver
Enter the coefficients of 6, 10 and -19, and solve

Write down the exact solutions

$$x_1 = \frac{-5 + \sqrt{139}}{6}$$
$$x_2 = \frac{-5 - \sqrt{139}}{6}$$

The question requests the answers to four significant figures
Find the decimal approximations to the solutions on your calculator

Round to four significant figures

$$x_1 = 1.132$$
$$x_2 = -2.798$$

Teacher only

Test your students on this in just a few clicks

- Create exam-aligned tests
- Tailor format & difficulty
- Quizzes or homework in minutes

Try Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources
Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard
Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company
About us
Exam Specificity

Subjects
Biology
Chemistry

Jobs
Terms
Privacy
Cookie Policy
Help and Support
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.